Ashmit Mukherjee

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New York University Abu Dhabi | Abu Dhabi, United Arab Emirates

Research Interests

Human-AI interaction; AI-mediated cooperation and collective decision-making; multi-agent systems; computational social science; rigorous evaluation of AI systems.

Education

New York University Abu Dhabi *Expected May 2027*
B.S. in Computer Science; minor in Economics
Study away: NYU Paris and NYU New York
Selected coursework: Natural Language Processing, Principles of Data Science, Statistics for Social and Behavioral Sciences, Data Structures and Algorithms, Operating Systems

Research Experience

Research Assistant, Human-AI Collaboration *Feb 2026 - present*
New York University Abu Dhabi | Ben Rosche and Hanan Salam

- Direct software implementation and review for faculty-led research on AI advice, coordination, and social dilemmas; translate study requirements into executable mechanics and data workflows.
- Coordinate development and review across Python/FastAPI, Empirica participant interfaces, session orchestration, and structured exports; contribute implementation-focused study-design feedback.
- Develop requirements for inspectable computational experiments, including validated actions, resumable execution, retained records, and independent checks of outcomes.
- Current computational validation uses deterministic fixture advisers; real-model experiments and the human-study platform have separate development and validation milestones.

Research Assistant, eBRAIN Lab *Feb 2026 - present*
New York University Abu Dhabi | Muhammad Shafique and Minghao Shao

- Extended BioInjectionEval to signal-peptide classification and cleavage-site prediction with ESM-2; contributed task adapters, model heads, and evaluation metrics.
- Implemented and evaluated biological-prior adaptation variants and comparisons with vanilla LoRA and full fine-tuning; ran multi-seed HPC studies across 150M, 650M, and 3B model scales.
- Examined prior and initialization controls, leakage checks, and low-resource settings. Recorded speed and memory comparisons alongside configurations, results, and limitations.
- Analyzed task-specific results under confirmatory controls, separating predictive improvements from evidence about the underlying biological mechanism.

Research Manuscript

Bio-LoRA: Biological Priors as Inductive Bias for Low-Rank Adaptation.
Co-first author. Unpublished research manuscript, 2026. Submitted to EMNLP 2026; not accepted.
OpenReview submission record | Research contribution and experimental scope

Independent Research

Synthetic Pilot Lab - Computer Science Capstone *Oct 2026 - present*

- Investigate whether synthetic participants can expose faults in experimental systems before human piloting, with reproducible diagnostic evidence and explicit evaluation criteria.
- Specify paired correct/faulty implementations, exposing and non-exposing scripted trajectories, and an independent evaluator that reconstructs expected state and payoffs.
- Established the standalone project and first diagnostic specification. Implementation and executed evaluations are the next milestone; model-based fault discovery remains planned work.

Selected Research and Engineering Projects

FieldLens - Evidence-linked Incident Review *Oct 2026*

- Directed the design and development of a local-model prototype using fictional incident records; specified bounded retrieval, application-owned claims, exact references, and deterministic failure handling.
- Separated output validity, proposition coverage, and relevance in saved synthetic comparisons; preserved incomplete answers, model failures, and portability limitations.

Hinglish Named-Entity Recognition *Fall 2024*

- Contributed training pipelines, evaluation tooling, and error analysis in a four-person course project using mBERT and XLM-RoBERTa on COMI-LINGUA.
- Encoder evaluation used the full 4,829-example test split; zero-shot language-model baselines used a separate 100-example sample. These evaluation scopes are distinct.

CAMP - Campus Asset Management Platform *Fall 2024*

- Built staff-workflow REST endpoints, equipment alerts, and dashboard statistics in a five-person project using React, Express, and MongoDB; configured backend integration testing and client API utilities.
- Served as Product Owner and Scrum Master in rotating team sprints.

Industry and Applied Experience

Machine Learning Engineer Intern, Zeek *Jun - Aug 2025*

- Built and evaluated customer-segmentation workflows; compared preprocessing and model choices and used SHAP to inspect model behavior.

Market Research Consultant, Global Media Alliance Broadcasting *Jun 2024*

- Conducted audience analysis, survey design, and customer-engagement research; presented quantitative and qualitative findings to senior leadership.

Leadership and Service

First Year Program Facilitator, NYU Abu Dhabi *Aug - Dec 2024*
Facilitated workshops, group activities, and weekly check-ins for more than 20 first-year students.
Front of House Staff, NYU Abu Dhabi Arts Center *Jan - May 2024*
Supported guest check-in, ticketing, and event operations for audiences exceeding 500 attendees.

Technical Skills

Programming: Python, JavaScript, C++, C#
ML and NLP: PyTorch, Transformers, scikit-learn, ESM-2, LoRA, XGBoost, SHAP
Research and data: Experimental design, model evaluation, pandas, NumPy, SciPy, statistical modeling, Matplotlib, Plotly
Systems: FastAPI, Empirica, REST APIs, Slurm, Docker, pytest, Git, GitHub Actions, MLflow, LaTeX